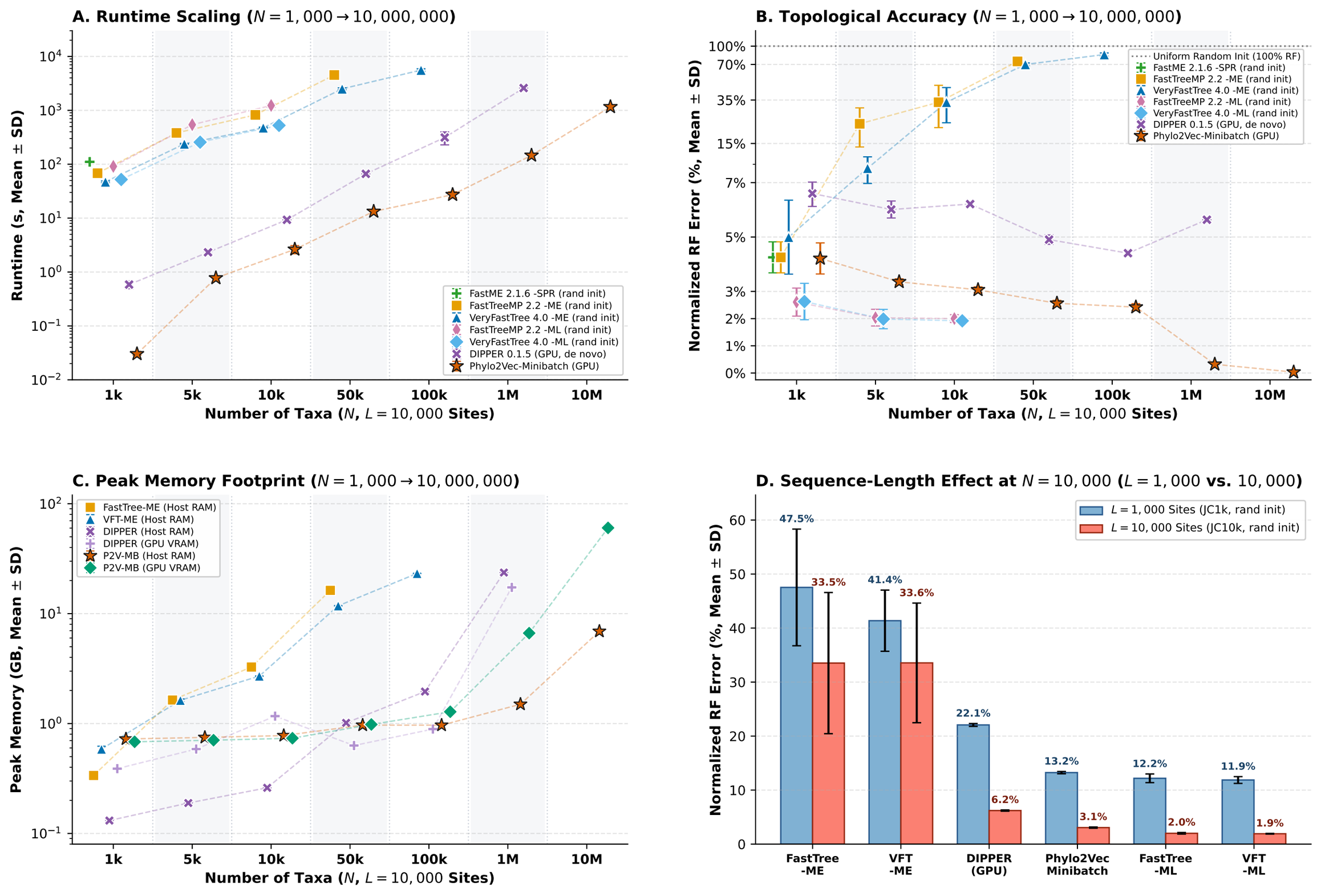

A. Runtime Scaling ($N = 1,000 \rightarrow 10,000,000$)
Runtime (s, Mean ± SD)
10^{4}
10^{3}
10^{2}
10^{1}
10^{0}
10^{-1}
10^{-2}
1k
5k
10k
50k
100k
1M
10M
Number of Taxa (N, $L = 10,000$ Sites)
FastME 2.1.6 -SPR (rand init)
FastTreeMP 2.2 -ME (rand init)
VeryFastTree 4.0 -ME (rand init)
FastTreeMP 2.2 -ML (rand init)
VeryFastTree 4.0 -ML (rand init)
DIPPER 0.1.5 (GPU, de novo)
Phylo2Vec-Minibatch (GPU)
B. Topological Accuracy ($N = 1,000 \rightarrow 10,000,000$)
Normalized RF Error (%, Mean ± SD)
100%
70%
35%
15%
7%
5%
3%
2%
1%
0%
Uniform Random Init (100% RF)
FastME 2.1.6 -SPR (rand init)
FastTreeMP 2.2 -ME (rand init)
VeryFastTree 4.0 -ME (rand init)
FastTreeMP 2.2 -ML (rand init)
VeryFastTree 4.0 -ML (rand init)
DIPPER 0.1.5 (GPU, de novo)
Phylo2Vec-Minibatch (GPU)
1k
5k
10k
50k
100k
1M
10M
Number of Taxa (N, $L = 10,000$ Sites)
C. Peak Memory Footprint ($N = 1,000 \rightarrow 10,000,000$)
Peak Memory (GB, Mean ± SD)
10^{2}
10^{1}
10^{0}
10^{-1}
FastTree-ME (Host RAM)
VFT-ME (Host RAM)
DIPPER (Host RAM)
DIPPER (GPU VRAM)
P2V-MB (Host RAM)
P2V-MB (GPU VRAM)
1k
5k
10k
50k
100k
1M
10M
Number of Taxa (N, $L = 10,000$ Sites)
D. Sequence-Length Effect at $N = 10,000$ ($L = 1,000$ vs. $10,000$)
Normalized RF Error (%, Mean ± SD)
60
50
40
30
20
10
0
$L = 1,000$ Sites (JC1k, rand init)
$L = 10,000$ Sites (JC10k, rand init)
47.5%
33.5%
41.4%
33.6%
22.1%
6.2%
13.2%
3.1%
12.2%
2.0%
11.9%
1.9%
FastTree -ME
VFT -ME
DIPPER (GPU)
Phylo2Vec Minibatch
FastTree -ML
VFT -ML